

Name:

Collaborators:

Instructions: Worksheets are graded mostly on completion and partially on correctness. Please write complete solutions showing explanations and key steps to the following problems, unless it says otherwise.

Use Eigenvalues to Analyze Equilibrium Stability

1. Linearizations of 1st-Order Non-Linear System of ODEs

Linearization is the process of approximating the solution behaviors nearby and around an equilibrium.

Consider this *non-linear homogeneous system of 1st-order ODEs*:

$$x' = 3x - \frac{7}{5}xy$$
$$y' = -y + \frac{4}{5}xy$$

with equilibrium solutions $(0,0)$ and $(\frac{5}{4}, \frac{15}{7})$.

- a. The following tasks determines the Jacobian matrix $J(x,y)$, and defines J_1 and J_2 which is $J(x,y)$ evaluated at the equilibrium solutions. Determining the Jacobian Matrix is an important step for linearizing a non-linear system of 1st-order ODEs.

Task	Show work here
Determine the partial derivatives $\frac{\partial f_1}{\partial x}$ and $\frac{\partial f_1}{\partial y}$ where $f_1 = 3x - \frac{7}{5}xy$.	
Determine the partial derivatives $\frac{\partial f_2}{\partial x}$ and $\frac{\partial f_2}{\partial y}$ where $f_2 = -y + \frac{4}{5}xy$.	
Construct the Jacobian matrix $J(x,y) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{bmatrix}$.	
Determine $J_1 = J(0,0)$.	
Determine $J_2 = J(\frac{5}{4}, \frac{15}{7})$.	

The *Jacobian Matrix* represents a tangent linear map that describes the local “slope” and directional behavior of an equilibrium solution.

2. Eigenvalues for Determining Equilibrium Stability

Consider the same non-linear system given in Problem (1) and the Jacobian matrices J_1 and J_2 .

The *eigenvalues* of the Jacobian matrix at an equilibrium can help us predict its stability because it represents the slope of the “straight-line” localized solutions around the equilibrium.

- a. The following tasks determines the eigenvalues of J_1 and uses them to determine the stability of the equilibrium $(0, 0)$.

Task	Show work here
Set $\det \left(J_1 - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0$.	
Write the characteristic equation with λ as the unknown.	
Solve for the two values of λ .	
Examine the properties of λ . Is it real, complex, or imaginary? Are they both positive, negative, or opposite signs?	
Note that J_1 represents the equilibrium $(0, 0)$. Use the properties of λ to determine the behavior of solutions around $(0, 0)$.	

- b. Do the same tasks in Part (a) to determine the eigenvalues of J_2 and use them to determine the stability of the equilibrium $\left(\frac{5}{4}, \frac{15}{7}\right)$.
- c. Draw solution curves around each equilibrium on the phase plane and observe their behaviors. Compare your observations to Parts (a) and (b).

